FIELD GUIDE

SwarmLoom

Watch a swarm of cursors weave it into being.

让一群光标，把它织出来。

SwarmLoom turns a page, a document or a model into a build you can watch. A small crowd of cursors takes the finished result apart into boxes, lines of text, paths and pictures, then puts it back together in reading order: frames first, then the words inside them, then the details.

Every cursor is an independent worker. When one finishes a task it picks the nearest open one, so the swarm spreads across the page and gathers where the work is. The timing comes from a seeded clock, which means the same file and the same seed always produce the same film, frame for frame.

One engine, many formats

Each format has a small adapter that reads the file and describes what to build. The engine never needs to know where a layer came from.

Format	Adapter reads	Build style
Markdown	remark AST, themed HTML	typed text
DOCX	paged HTML	typed text, swept images
PDF	pdf.js page bitmap	typed line crops
SVG	path geometry	pen and trace
Slides	positioned frames	boxes, then text
3D model	WebGL patches	vertices, edges, faces

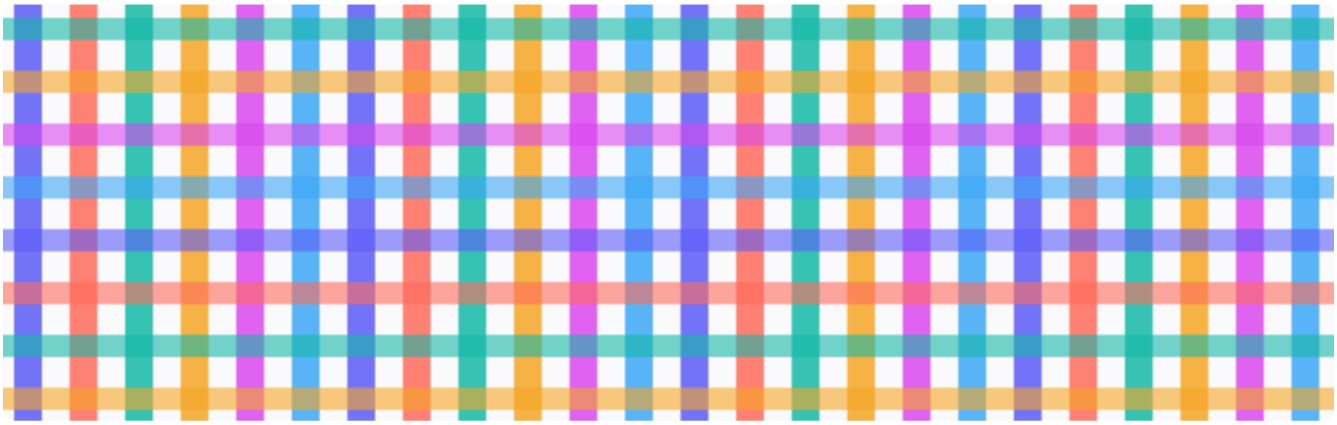


Figure 1. A woven test pattern drawn for this guide.

Build speed by format

Median layers revealed per second on a mid-range laptop at speed 1. Text-heavy formats run slower because every line is typed; vector formats are drawn stroke by stroke.

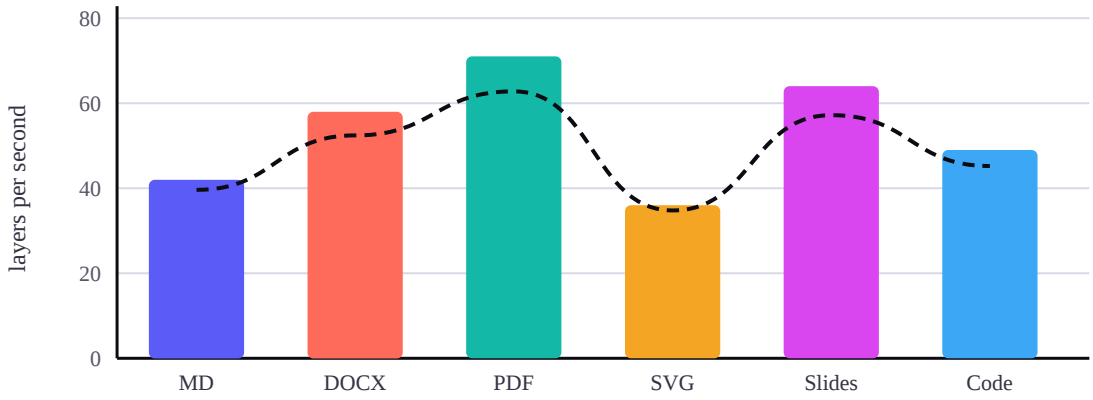


Figure 2. Bars show the median; the dashed line shows the trend at speed 2.

Notes for authors

Keep headings short. The swarm types them at a larger size and lets them grow as they are written, so a long heading takes a while to land.

Tables are drawn as a grid first, then filled row by row. Very wide tables still work, but they are easier to follow when each cell holds a few words.

Pictures are swept in from top to bottom once their frame is placed. Large photos look best when they have a little space around them.

Exported films keep the sound in step with the picture, because both come from the same list of events recorded during the build.

SwarmLoom sample document. Written for the SwarmLoom repository and released under the MIT license.